

Analytic Functions of a Complex Variable

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Handwritten lecture notes, transcribed

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(Analytic functions of a complex variable)

Note – (Mod 01 – lec 1) of this topic is written in lec 25 of Quantum mechanics; it is transcribed below, ahead of Lecture 2. The group-theory material that shares those same notebook pages is collected in the section “Group Theory: $SO(3)$ and $SU(2)$ ” of the Classical Mechanics notes.

1 Module 1, Lecture 1

(Mod-01. lec-01, analytic functions of complex variable)

Some mathematical prerequisite :-

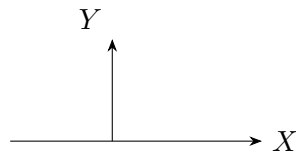
- (1) complex numbers
- (2) Equations to curves in the plane in terms of z and z^*
- (3) The Riemann sphere and stereographic projection
- (4) Analytic functions of z and the cauchy Riemann conditions.
- (5) The real and imaginary parts of an analytic function.

complex variable $z = x + iy$, ($x, y \in \mathbb{R}$) and $z^* = x - iy$.

we will talk more on analytic functions of z i.e. $x + iy$.

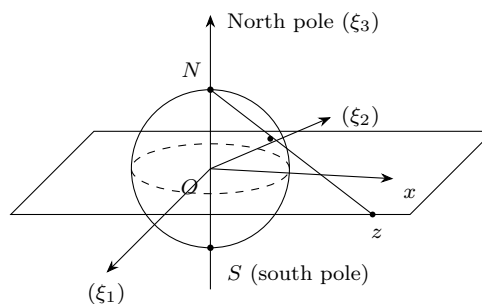
analytic function is purely a function of ‘ $x + iy$ ’ and does not involve ‘ $x - iy$ ’. (z^*).

stereographic projection :-



The complex plane extends to infinity in infinite possible directions, so we gonna have infinite number of complex variables at $(x, y \rightarrow \infty)$. So to get one point at infinity, the whole x - y complex plane is compacted to sphere of unit radius.

Riemann sphere



($R = 1$). let the points on Coo sphere be (ξ_1, ξ_2, ξ_3) , so that

$$\begin{aligned}\xi_1 &= \sin \theta \cos \phi, & \xi_2 &= \sin \theta \sin \phi, & \xi_3 &= \cos \theta \\ (\xi_1^2 + \xi_2^2 + \xi_3^2 &= 1)\end{aligned}$$

we draw any line from North pole to complex plane, then this line intersects at one point on sphere (ξ_1, ξ_2, ξ_3) and same line intersects $((x, y) \equiv z)$ complex plane.

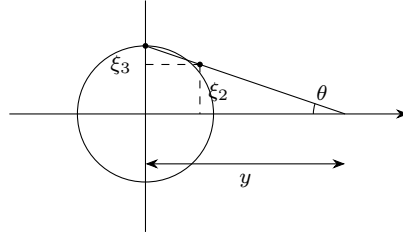
so $\exists (\xi_1, \xi_2, \xi_3)$ for every complex no. on plane.

All the points for which $|z| > 1$ are mapped from Northan hemisphere points and remaining points of complex plane $|z| < 1$ are mapped from southern hemisphere.

$|z| = 1$ is mapped from unit circle on this plane.

$|z| = 0$ is mapped from south pole in the Riemann sphere.

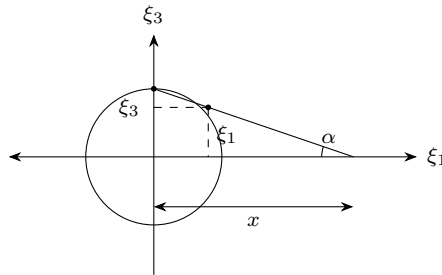
let us look it in ξ_2 - ξ_3 plane :-



By similarity of triangles

$$\begin{aligned}\tan \theta &= \frac{1}{y} = \frac{1 - \xi_3}{\xi_2} = \frac{\xi_3}{y - \xi_2} \\ \Rightarrow y &= \frac{\xi_2}{1 - \xi_3}\end{aligned}$$

let us look from ξ_1 - ξ_3 plane :-



$$\begin{aligned}\tan \alpha &= \frac{1}{x} = \frac{1 - \xi_3}{\xi_1} = \frac{\xi_3}{x - \xi_1} \\ \left(x &= \frac{\xi_1}{1 - \xi_3} \right)\end{aligned}$$

(similarly, $z \neq \frac{\xi_3}{1 - \xi_3}$ (from ξ_1 - ξ_2 plane), HaHa we do not have z coordinate on x - y plane).

we know that, $\xi_1 = \sin \theta \cos \phi$, $\xi_2 = \sin \theta \sin \phi$, $\xi_3 = \cos \theta$, so

$$x = \frac{\xi_1}{1 - \xi_3} = \frac{\sin \theta \cos \phi}{1 - \cos \theta} = \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} \cos \phi}{2 \sin^2 \frac{\theta}{2}} = \cot \frac{\theta}{2} \cos \phi$$

$$y = \frac{\xi_2}{1 - \xi_3} = \frac{\sin \theta \sin \phi}{1 - \cos \theta} = \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} \sin \phi}{2 \sin^2 \frac{\theta}{2}} = \cot \frac{\theta}{2} \sin \phi$$

$$z = x + iy = \cot \frac{\theta}{2} (\cos \phi + i \sin \phi)$$

$$\boxed{z = \cot \frac{\theta}{2} e^{i\phi}}$$

$$z = x + iy = \frac{\xi_1 + i\xi_2}{1 - \xi_3}, \quad z^* = \frac{\xi_1 - i\xi_2}{1 - \xi_3}$$

But we need (ξ_1, ξ_2, ξ_3) for x, y or $z = x + iy$

$$\begin{aligned} \xi_1 &= \frac{2x}{x^2 + y^2 + 1} = \frac{z + z^*}{|z|^2 + 1} \\ \xi_2 &= \frac{2y}{x^2 + y^2 + 1} = \frac{z - z^*}{i(|z|^2 + 1)} \\ \xi_3 &= \frac{x^2 + y^2 - 1}{x^2 + y^2 + 1} = \frac{|z|^2 - 1}{|z|^2 + 1} \end{aligned}$$

ex : $x, y = 0$ gives $\xi_1 = 0$, $\xi_2 = 0$, $\xi_3 = -1$, i.e. $(0, 0, -1) \rightarrow$ south pole.

The point at infinity and the extended complex plane

All points at ∞ from all directions are going to map to North pole. So we can call 'N' the point at infinity. we can reach infinity from any direction but that is going to mapped to North pole.

so complex plane without infinity is called just complex plane, $\mathbb{C} : \{|z| < \infty\}$.

if we include infinity i.e. 'North pole', it is called extended complex plane, $\hat{\mathbb{C}}$ = extended complex plane $\{|z| \leq \infty\}$.

so by compacting the complex plane to Riemann sphere we can perform calculus on it, without worrying about infinity. as this Riemann sphere includes a single point which represents infinity.

(Q) Do we have notion of distance between two points on this Riemann sphere ?

One way is to assume first point at North pole and the distance along longitude to the second point, and that becomes geodesic distance / shortest distance on surface of sphere.

second method :-

If we have two points z_1, z_2 in complex plane then $|z_1 - z_2|$ is distance between two points in this complex plane.

So what is corresponding distance on this sphere ? It will be the chordal distance between two points on the sphere.

chordal distance

$$d(z_1, z_2) = \frac{2|z_1 - z_2|}{\sqrt{(|z_1|^2 + 1)(|z_2|^2 + 1)}}$$

consequences :-

- (1) $d(z_1, z_2) = 0$ iff $z_1 = z_2$
- (2) $d(z_1, z_2) \geq 0$ $\therefore$ it can not be negative
- (3) $d(z_1, z_3) \leq d(z_1, z_2) + d(z_2, z_3)$ (Triangle inequality)
- (4) $d(z_2, z_1) = d(z_1, z_2)$

$$\begin{aligned} d(z_1, \infty) &= \lim_{|z_2| \rightarrow \infty} \frac{2|z_1 - z_2|}{\sqrt{|z_1|^2 + 1} \left(\sqrt{|z_2|^2 + 1} \right)} \\ &\Rightarrow \frac{2}{\sqrt{|z_1|^2 + 1}} \end{aligned}$$

so distance between origin and point at ' ∞ ' is $d(0, \infty) = \frac{2}{\sqrt{0+1}} = 2$.

$z_1 = 0$ is represented on sphere by $(0, 0, -1) \rightarrow$ south pole.

$z_2 \rightarrow \infty$ is represented on sphere by $(0, 0, 1) \rightarrow$ North pole.

So, $d(0, \infty) = 2$.

Analytic functions :-

$f(z)$ is analytic in some region in this complex plane if it satisfies couple of relations.

$f(z) = u(x, y) + i v(x, y) \downarrow$ analytic iff cauchy Riemann conditions are satisfied :

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

These Cauchy–Riemann conditions really means that the $f(z)$ is not function of any linear combination of x and y but only of $(x + iy)$.

All these combinations are excluded to be called as analytic function – other combinations (linear) of x, y :

$$x + y, \quad 2x + y, \quad x - y, \quad x - iy, \quad ax + iby \quad (a, b \in \mathbb{R}) \quad (a \neq 1, b \neq 1), \quad 2x + iy$$

or $\left(\frac{\partial f}{\partial z^*} = 0 \right)$ is sufficient & necessary condition to be satisfied for ' f ' to be analytic function,

since $\frac{\partial f}{\partial z^*} = \frac{\partial f}{\partial x} + \frac{\partial f}{(-i \partial y)}$.

$$\left. \begin{aligned} x &= \frac{z + z^*}{2} \\ y &= \frac{z - z^*}{2i} \end{aligned} \right\} \quad \frac{\partial f}{\partial z^*} = \frac{\partial f}{\partial x} + i \frac{\partial f}{\partial y} = 0$$

$$\begin{aligned}
f &= u + iv \\
\frac{\partial f}{\partial z^*} &= \left(\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \right) + i \left(\frac{\partial u}{\partial y} + i \frac{\partial v}{\partial y} \right) = 0 \\
&\quad \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) + i \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) = 0 = 0 + 0i \\
\Rightarrow \quad \frac{\partial u}{\partial x} &= \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} \quad \} \text{ cauchy Riemann condition.}
\end{aligned}$$

(Q) Are these analytic function.

$$\begin{array}{llll}
\text{(a) } x & \text{(b) } y & \text{(c) } (2x + iy) & \text{(d) } x - iy \\
\text{(e) } re^{i\theta} & \text{(f) } \theta & \text{(h) } 2x + 3iy &
\end{array}$$

Ans

$$\text{(a) } f(z) = x = \frac{z + z^*}{2}, \quad \frac{\partial f}{\partial z^*} \neq 0 \text{ so } \underline{\text{No}}, \text{ not analytic function.}$$

$$\text{(b) } f(z) = y = \frac{z - z^*}{2i}, \quad \frac{\partial f}{\partial z^*} \neq 0 \text{ so } \underline{\text{No}}.$$

$$\text{(c) } f(z) = (2x + iy)$$

$$\begin{aligned}
u &= 2x & u_x &= 2 & u_y &= 0 \\
v &= y & v_x &= 0 & v_y &= 1
\end{aligned}$$

$$(u_x \neq v_y) \quad \underline{\text{No}}.$$

$$\text{(d) } f(z) = x - iy = z^*, \quad \frac{\partial f}{\partial z^*} = 1 \neq 0 \text{ so } \underline{\text{No}}.$$

$$\text{(e) } f(z) = re^{i\theta}, \text{ where } r = \sqrt{x^2 + y^2} \text{ and } \theta = \tan^{-1}(y/x)$$

$$\begin{aligned}
z &= x + iy, \quad z^* = x - iy, \quad zz^* = x^2 + y^2 \\
f(z) &= (zz^*)^{1/2} e^{i \tan^{-1} y/x}
\end{aligned}$$

$$\frac{\partial f}{\partial z^*} \neq 0, \text{ so } \underline{\text{No}}.$$

$$\text{(f) } f(z) = \theta$$

$$z = re^{i\theta}, \quad z^* = re^{-i\theta}, \quad \frac{z}{z^*} = e^{2i\theta}$$

$$\ln \left(\frac{z}{z^*} \right) = 2i\theta, \quad \theta = \frac{1}{2i} \ln \left(\frac{z}{z^*} \right)$$

$$\text{so } f(z) = \frac{1}{2i} \ln \left(\frac{z}{z^*} \right) = \theta \text{ and } \frac{\partial f}{\partial z^*} \neq 0 \text{ so } \underline{\text{No}}.$$

$$\text{(h) } f = 2x + 3iy$$

$$f = 3(x + iy) - x = 3z - \left(\frac{z + z^*}{2} \right) = \frac{5z - z^*}{2}, \quad \frac{\partial f}{\partial z^*} \neq 0, \text{ so not analytic}$$

harmonic functions

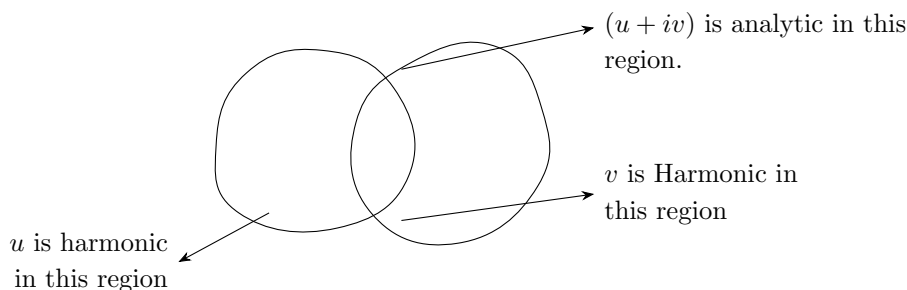
In the region in which $f(z)$ is analytic the real part and imaginary part of $f(z)$ separately satisfies Laplace equations. i.e.

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 = \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}$$

$$f(z) = u(x, y) + i v(x, y)$$

(so the functions which satisfies laplace equations are called as harmonic functions, so real and imaginary parts of analytic function are harmonic functions.)

we can find imaginary part if real part is provided and the region in which function is analytic is provided.



entire function

If $f(z)$ is analytic in whole complex plane ($|z| < \infty$) then $f(z)$ is called as entire function.

ex

- (1) $f(z) = z$ is an entire function
- (2) $f(z) = z^2$ – yes, entire function
- (3) $f(z) = z^n$, $n = +ve$ integer – Yes, entire function
- (4) $f(z) = P_n(z) \leftrightarrow$ some polynomial of degree n – yes, entire function
- (5) $f(z) = e^{\pm z} = \left(1 + \frac{z}{1} + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots\right)$ true for all $|z| < \infty$. Yes, entire function.
- (6) $f(z) = \frac{e^z + e^{-z}}{2} = \cosh z$ – Yes, entire fun.
- (7) $f(z) = \sinh z = \frac{e^z - e^{-z}}{2}$ – yes, entire function
- (8) Both are analytic & entire function :

$$f(z) = \cos z = \frac{e^{iz} + e^{-iz}}{2}, \quad f(z) = \sin z = \frac{e^{iz} - e^{-iz}}{2i}$$

- (9) $f(z) = \tan z = \frac{\sin z}{\cos z} = \frac{e^{iz} - e^{-iz}}{(e^{iz} + e^{-iz})i}$ – Not clear at all as function blows up at some point.

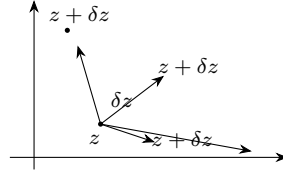
2 Module 1, Lecture 2

outline :-

- (1) The derivative of an analytic function
- (2) power series as analytic functions
- (3) convergence of power series.

(1) Derivative of analytic function :-

let $f(z)$ be analytic function in complex plane $\bar{z}$.



$$\frac{df}{dz} = \lim_{\delta z \rightarrow 0} \frac{f(z + \delta z) - f(z)}{\delta z}$$

Here we can see that we do not know in which direction we should proceed to calculate derivative?

let us take a general case

$$\delta z = \epsilon e^{i\alpha}, \quad z = x + iy$$

$$\begin{aligned} \frac{df}{dz} &= \lim_{\delta z \rightarrow 0} \frac{f(z + \delta z) - f(z)}{\delta z} = \\ &= e^{-i\alpha} \left\{ \left(\frac{\partial u}{\partial x} \cos \alpha + i \frac{\partial v}{\partial y} \sin \alpha \right) - i \left(\frac{\partial u}{\partial y} \cos \alpha - i \frac{\partial v}{\partial x} \sin \alpha \right) \right\} \end{aligned}$$

What direction should α be.

One way to define the derivative as it is same in all directions specified by ' α ', i.e. it is independent of α .

$$\frac{df}{dz} = \text{independent of } \alpha \quad \text{when} \quad \left. \begin{array}{l} u_x = v_y \\ u_y = -v_x \end{array} \right\} \underline{\text{CR Relations}}$$

i.e. For analytic function their derivative does not depend on direction specified by angle α .

Some Real functions are not infinitely differentiable.

Ex $f(x) = x^2$ (only twice differentiable)

$$\begin{aligned} f'(x) &= 2x \\ f''(x) &= 2 \\ f'''(x) &= 0 \end{aligned}$$

$\therefore$ and so does $f^n(x)$ does not exist.

In complex functions we do not have such complications.

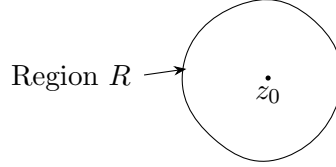
If $f(z)$ is analytic $\Rightarrow f'(z)$ is analytic $\Rightarrow f''(z)$ is analytic

The derivative of analytic function is also analytic function so does $f''(z)$ & higher derivatives.

so $f(z)$ is analytic ($\Leftrightarrow$) all its derivative are also analytic.

Power series representation :-

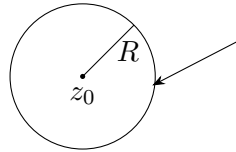
We can always represent a given analytic function in some power series about some point z_0 which lies inside some region of analyticity / region of convergence.



$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n \quad (\text{Taylor series})$$

$$\left(a_n = \frac{1}{n!} \frac{\partial^n f(z)}{\partial z^n} \Big|_{z=z_0} \right) \quad (\text{Inverse of Taylor series})$$

But this power series should be convergent to some finite value within some region of analyticity / circle of convergence about point z_0 with radius R . such that



$$\left(\sum_{n=0}^{\infty} |a_n (z - z_0)^n| < \infty \right)$$

To find radius of convergence $|z - z_0|$ we need,

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1} (z - z_0)^{n+1}}{a_n (z - z_0)^n} \right| < 1$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| |z - z_0| < 1$$

$$\left(R < \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| \right)$$



radius of convergence.

Now there is no guarantee that this $\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right|$ exists

So we define

$$\left(R = \limsup_{n \rightarrow \infty} |a_n|^{1/n} \right)$$

If radius of convergence for a given $f(z)$ is ' ∞ ' then this function is entire function (defined at ∞ too).

Examples:

(1) $f(z) = e^z$

$$e^z = \sum_n \frac{z^n}{n!}$$

to converge

$$\lim_{n \rightarrow \infty} \left| \frac{z^{n+1}}{(n+1)!} \left(\frac{n!}{z^n} \right) \right| < 1$$

$$\Rightarrow \lim_{n \rightarrow \infty} \left| \frac{z}{n+1} \right| < 1$$

for all finite value of z , ($z < \infty$)

$$\lim_{n \rightarrow \infty} \left| \frac{z}{n+1} \right| \rightarrow 0 < 1 \quad (\text{converges}).$$

(Radius of convergence) :-

$$R \leq \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right|$$

$$R \leq \lim_{n \rightarrow \infty} \left| \frac{z^n (n+1)!}{n!} \right|$$

$(R \rightarrow \infty) \quad (\text{entire function}).$

(2) $f(z) = 1 + z + z^2 + z^3 + \dots \infty = \sum_{n=0}^{\infty} z^n$

Is this function entire function?

Is the series convergent, where does this series converge?

$f(z)$ converges for $|z| < 1$.

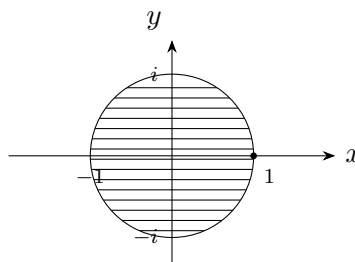
we can see

$$\lim_{n \rightarrow \infty} \left| \frac{z^{n+1}}{z^n} \right| < 1$$

$$\left(\lim_{n \rightarrow \infty} |z| < 1 \right)$$

So Radius of convergence should be $R = 1$.

check, $R \leq \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| \Rightarrow (R = 1) \quad (\text{as } a_n = a_{n+1} = 1).$



Inside this region series $\sum_{n=0}^{\infty} z^n$ is convergent.

$$f(z) = \sum_{n=0}^{\infty} z^n \quad \text{diverges for all } z \text{ such that } |z| > 1$$

Inside region of convergence / analyticity :

we can identify that

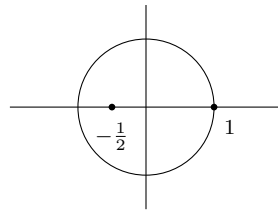
$$f(z) = 1 + z + z^2 + \dots = \frac{1}{1-z}$$

$$f(z) = \begin{cases} \sum_{n=0}^{\infty} z^n & (|z| < 1) \\ \frac{1}{1-z} & (z \neq 1) \text{ \& } (|z| < 1) \end{cases}$$

so the representation of $f(z) = \frac{1}{1-z}$ matches with $f(z) = \sum_{n=0}^{\infty} z^n$ point by point within region $(|z| < 1)$ or $|z - 0| < 1$.

But $g(z) = \frac{1}{1-z}$ is making sense / definable at all z . except $z = 1$. So, $g(z) = \frac{1}{1-z}$ is called as analytic continuation of $f(z)$.

writing power series $f(z) = 1 + z + z^2 + \dots$ about $(z_0 = -\frac{1}{2})$.



within region $|z| < 1$

$$\begin{aligned} f(z) &= \frac{1}{1-z} \\ &= \frac{1}{1 - (z - (-\frac{1}{2})) + \frac{1}{2}} \\ f(z) &= \frac{1}{1-z} = \frac{1}{1 + \frac{1}{2} - \frac{1}{2} - z} = \frac{1}{\frac{3}{2} - (z + \frac{1}{2})} = \frac{1}{\frac{3}{2} (1 - \frac{2}{3} (z + \frac{1}{2}))} \\ f(z) &= \frac{2}{3} \left(1 - \frac{2}{3} \left(z + \frac{1}{2} \right) \right)^{-1} \end{aligned}$$

$$((1-x)^{-1} = 1 + x + x^2 + \dots)$$

$$f(z) = \frac{2}{3} \sum_{n=0}^{\infty} \left(\frac{2}{3} \right)^n \left(z + \frac{1}{2} \right)^n$$

$\hookrightarrow$ power series about $(z_0 = -\frac{1}{2})$

where does this series converge.?

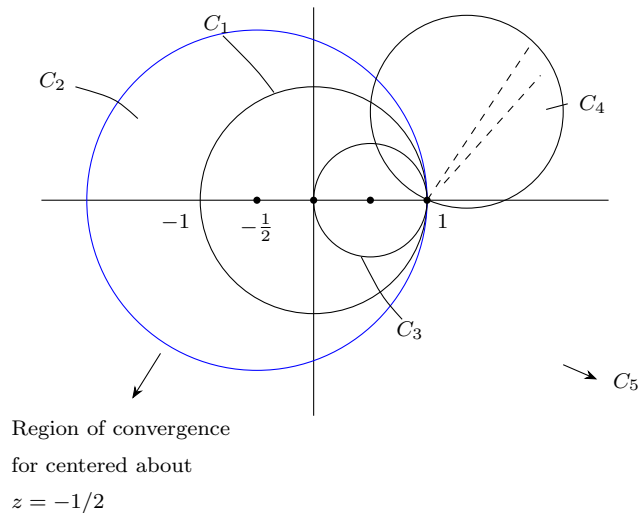
for convergence

$$\lim_{n \rightarrow \infty} \left| \frac{\left(z + \frac{1}{2}\right)^{n+1} \left(\frac{2}{3}\right)^{n+1}}{\left(z + \frac{1}{2}\right)^n \left(\frac{2}{3}\right)^n} \right| < 1$$

$$\lim_{n \rightarrow \infty} \left| \left(z + \frac{1}{2}\right) \left(\frac{2}{3}\right) \right| < 1$$

$$\left(\left| z + \frac{1}{2} \right| < \frac{3}{2} \right)$$

$\hookrightarrow$ circle of radius $3/2$ centered at $(z = -1/2)$



We can write the power series

$$f(z) = 1 + z + z^2 + \dots \infty$$

		Region of convergence / circle of converg[?]	
centered about	$z = 0$	$\longrightarrow$	$C_1 : z < 1$
	$z = -1/2$	$\longrightarrow$	$C_2 : \left z + \frac{1}{2} \right < \frac{3}{2}$
	$z = 1/2$	$\longrightarrow$	$C_3 : \left z - \frac{1}{2} \right < \frac{1}{2}$
	$z = 3/2 + i$	$\longrightarrow$	$C_4 : \left z - \left(\frac{3}{2} + i\right) \right < \frac{\sqrt{5}}{2}$
	$z = 2 - i$	$\longrightarrow$	$C_5 : z - (2 - i) < \sqrt{5}$

All these representations are valid for given power series but they are valid in given Region of analyticity or convergence given by region C_i , $1 \leq i \leq 5$.

Note :: So A given $f(z)$ can have infinite no. of representations. valid in different regions.

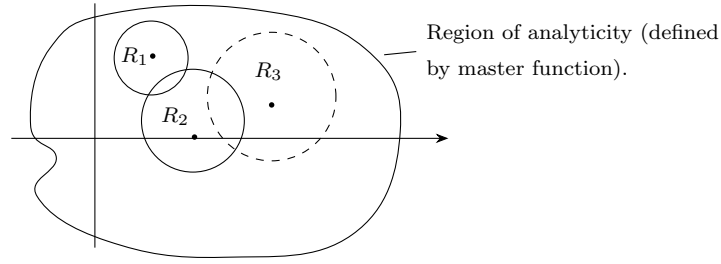
All these regions / representations are analytic continuations of each other.

We may or may not be able to find master representa-tions from which you can find other representations For $f(z) = 1 + z + z^2 + \dots$ (master representation is $f(z) = \frac{1}{1-z}$). A given representation of function is valid in some region of convergence so power series diverges outside region of convergence & converges for all 'z' inside circle of convergence. We are not sure about behaviour of $f(z)$ at boundary of convergence.

Region of convergence $\in$ Region of analyticity.

Region of convergence $<$ Region of analyticity.

$f(z) = 1 + z + z^2 + \dots \infty$ is only valid within region $(|z| < 1)$.



R_1 & R_2 , R_3 are regions of convergence of given power series about some different z_{01}, z_{02}, z_{03} .

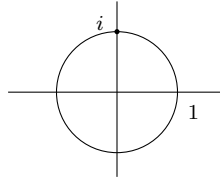
Behaviour at Boundary :-

(1) On boundary of circle of convergence., The series provides some use. (If series oscillates).

for, $f(z) = 1 + z + z^2 + z^3 + \dots$

$$f(z) = \frac{1}{1-z} \quad (\text{master representation})$$

at $(z = 1)$ $f(z) \rightarrow \infty$ $\left(\begin{array}{c} \text{singularity} \\ \text{at } z = 1 \end{array} \right)$



(a) $f(-1) = 1 - 1 + 1 - 1 + 1 - 1 + \dots$ ✓ Cisaró sum

lets take average of outcomes.

Possible outcomes = 1, 0

$$\text{avg} = \frac{1+0}{2} = 1/2$$

$$f(-1) = 1/2$$

we can check via $f(z) = \frac{1}{1-z} \Big|_{z=-1} = \frac{1}{2}$

(b) $f(i) = 1 + i + i^2 + i^3 + i^4 + \dots$ ✓ Cisaró sum.

$$= 1 + i - 1 - i + \underline{1} + \dots$$

Possible outcomes = 1, $1+i$, i , 0,

$$f(i) = \text{avg}(1, 1+i, i, 0) = \frac{2(1+i)}{4} = \frac{i+1}{2}$$

$$f(i) = \frac{1}{1-i} = \frac{1+i}{1-i^2} = \frac{1+i}{2}$$

Theorem : Any power series representation of $f(z)$ must have at least one singularity on circle of convergence.

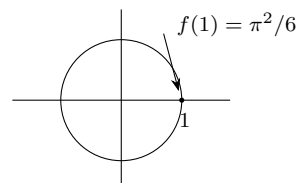
(2) Convergence / absolute convergence is also possible for z which lies on boundary of circle of convergence.

Ex $f(z) = \sum_{n=p}^{\infty} \frac{z^n}{n^2}$ converges for $\left| \frac{z^{n+1}}{(n+1)^2} \frac{n^2}{z^n} \right| < 1$

$$\lim_{n \rightarrow \infty} \left| \frac{z n^2}{(n+1)^2} \right| < 1$$

converges for. $|z| < 1$

In fact it converges even for points $|z| = 1$. too.



But $\exists$ a point on boundary $|z| = 1$ such that It has a singularity.

(In fact $f(z)$ is singular at $z = 1$.)

If we have $g(z) = (z-1) \ln(z-1)$

$$\left(\lim_{z \rightarrow 1} g(z) = 0 \right)$$

as $(z \rightarrow 1) \quad g(z) \rightarrow 0$

as log is weaker than power

$\log(z-1)$ is weaker than $(z-1)$.

Singular part is zero. But it is singular at $z = 1$

Singularity does not necessarily mean $f(z) \rightarrow \infty$, it is singular in sense of theory of analytic functions.

(Ex 2) $f(z) = z + z^2 + z^4 + z^8 + \dots$

$$f(z) = \sum_{n=0}^{\infty} z^{2^n} < \infty \text{ for } \left| \frac{z^{2^{n+1}}}{z^{2^n}} \right| < 1$$

$$\Rightarrow \lim_{n \rightarrow \infty} \left| \frac{z^{2^n} \cdot z^{2^n}}{z^{2^n}} \right| < 1$$

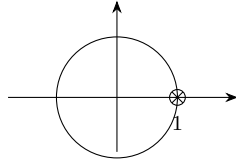
(Radius of convergence = 1)

$$\lim_{n \rightarrow \infty} |z^{2^n}| < 1$$

only if $|z| < 1$

for $(z = 1) f(z) = 1 + 1 + 1 \dots$ diverges.

$(f(z) \text{ is singular at } z = 1)$



we can also write

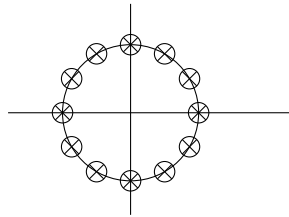
$$f(z) = z + f(z^2)$$

$$f(z^2) = z + z^2 + z^3 + \dots$$

In this, we have singularity at $z = \pm 1$

Similarly $f(z) = z + z^2 + f(z^4)$

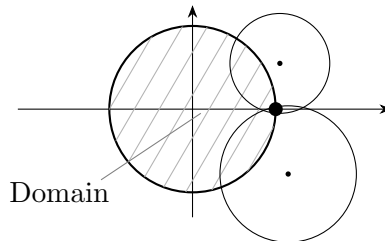
($\rightarrow$ have more singular points.



In this way $|z| = 1$ forms natural boundary where it is singular at infinite points on boundary $|z| = 1$.

So for $f(z) = z + z^2 + z^4 + z^8 + \dots$

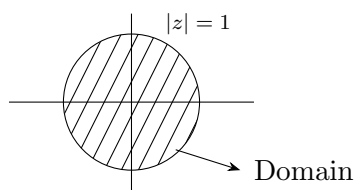
we can't have any other representation except as shown in below fig. for $f(z)$.



There is no analytic function continuation possible beyond the natural boundary due to presence of infinite singular points on $|z| = 1$.

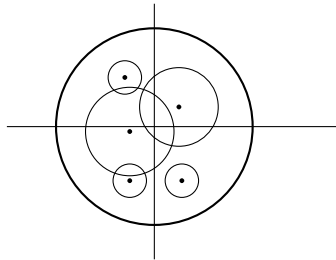
$$f(z) = \sum_{n=0}^{\infty} z^{n!} \quad \text{converges for} \quad \left| \frac{z^{(n+1)!}}{z^{n!}} \right| < 1$$

$$|z| < 1$$



for this $f(z)$ too we have infinite no of singular points on $|z| = 1$, so we only have this one representation of $f(z)$.

so $\exists$ such $f(z)$ for which we can only have one power series expansion.

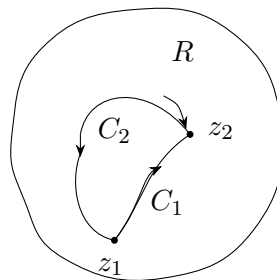


Inside we can have many representations.

3 Module 1, Lecture 3: Calculus of Residues

We have seen that if a function $f(z)$ is analytic in some region, then C-R conditions are satisfied, derivative is uniquely defined at every point, derivative itself is an analytic function & $f(z)$ is infinitely differentiable.

Integral of analytic function $f(z)$:-



$\left(\int_{z_1}^{z_2} f(z) dz \right)$ is independent of path 'C'.

It is similar to line integral of $\nabla\phi(r, t)$.

$$\int_a^b \nabla\phi \cdot d\vec{r} = \phi(b) - \phi(a)$$

(Cauchy's Integral theorem):

So $I = \oint f(z) dz = 0$ (Cauchy's Integral theorem)

$$\begin{aligned} \text{as } I &= \int_{z_1(C_1)}^{z_2} f(z) dz + \int_{z_2(C_2)}^{z_1} f(z) dz \\ &= \int_{z_1(C_1)}^{z_2} f(z) dz - \int_{z_1(C_2)}^{z_2} f(z) dz \\ &= 0 \quad \text{as } \int_{z_1(C_1)}^{z_2} f(z) dz = \int_{z_1(C_2)}^{z_2} f(z) dz. \end{aligned}$$

we can actually deform the path / contour of integration without changing its value as long as path remains in analytic region.

Singularity :

(A) Removable singularities :- upon proper redefinition of function this singularity can be removed.

example :

$$(1) \quad f(z) = \frac{\sin z}{z}$$

at $z = 0$ $f(z)$ is having removable singularity

$$\text{as } \lim_{z \rightarrow 0} f(z) = \lim_{z \rightarrow 0} \frac{\sin z}{z} = \lim_{z \rightarrow 0} \frac{\cos z}{1} = 1$$

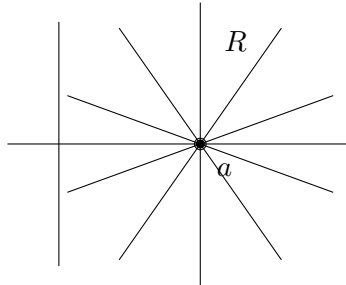
$$\text{If we redefine, } f(z) = \begin{cases} \frac{\sin z}{z} & z \neq 0 \\ 1 & z = 0 \end{cases}$$

this function is now free from singularity.

(B) Simple pole :-

If a function $f(z)$ is not analytic at $z = a$ and analytic everywhere / or around it for some radius of convergence R . then

(R may go to ∞).



Then we can express such $f(z)$

$$f(z) = \underbrace{\frac{C_{-1}}{(z-a)}}_{\text{singular part}} + \underbrace{\sum_{n=0}^{\infty} C_n (z-a)^n}_{\text{Regular part}}$$

$f(z)$ is said to have simple pole at $z = a$.

$$f(z) = \frac{\boxed{c_{-1}}}{(z-a)^{(1)}} + \sum_{n=0}^{\infty} c_n (z-a)^n$$

→ Residue (pointing to c_{-1}); ↓ Simple pole at $z = a$ (pointing to $(z-a)^{(1)}$)

and, residue at the pole $z = a$ is c_{-1}

Singularity may not be always obvious.

example

$$(1) f(z) = \frac{\sin z}{z^2}$$

$$= \frac{\left(z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots \right)}{z^2}$$

$$= \left(\frac{1}{z} \right) + \left\{ -\frac{z}{3!} + \frac{z^3}{5!} - \dots \right\}$$

↓ Singular part (pointing to $\frac{1}{z}$); $\underbrace{\hspace{1cm}}$ Regular part (under the braced terms)

Simple pole at $z = 0$

with residue = 1

$$(2) f(z) = \frac{g(z)}{h(z)}$$

lets say, $g(a) \neq 0$ & $h(a) = 0$

Then,

$$f(z) = \frac{g(a) + (z-a)g'(a) + \frac{(z-a)^2}{2!}g''(a) + \dots}{\overset{0}{h(a) + (z-a)h'(a) + \dots}}$$

$$= \lim_{z \rightarrow a} \frac{g(a) + (z-a)g'(a) + \dots}{(z-a)h'(a) + \dots}$$

as $\lim z \rightarrow a$

$$g(z) \simeq g(a)$$

$$h(z) \simeq (z-a)h'(a)$$

$$f(z) \simeq \frac{g(a)}{h'(a)} \frac{1}{(z-a)}$$

→ simple pole at $z = a$

with residue $\frac{g(a)}{h'(a)}$

$$\left\{ \begin{array}{l} \text{In general, for Simple pole at } z = a \\ \text{The residue} = \lim_{z \rightarrow a} (z-a) f(z) \end{array} \right\}$$

as

$$f(z) = \frac{c_{-1}}{(z-a)} + \sum_{n=0}^{\infty} c_n (z-a)^n$$

$$(z-a)f(z) = c_{-1} + \sum_{n=0}^{\infty} c_n (z-a)^{n+1}$$

$$\lim_{z \rightarrow a} (z-a)f(z) = c_{-1} = \text{Residue at simple pole.}$$

$$(3) f(z) = \frac{1}{\sin \pi z} \quad \text{Find } z \text{ for which } f(z) \text{ is singular and find Residue at each pole.}$$

$f(z)$ is singular for $\pi z = (\text{Integer}) \pi$

or $z = \text{integer}$ (all integers)

or $z = n$, $n = \text{Integer}$

$f(z)$ has simple poles at all integers.

as $z \rightarrow 0$, $\sin \pi z \rightarrow (\pi z)$ as $\left(\sin \pi z = \pi z - \frac{(\pi z)^3}{3!} + \dots \right)$

so as $z \rightarrow 0$, $f(z) \cong \frac{1}{\pi z}$

so simple pole at $(z = 0)$ with residue $\left(\frac{1}{\pi} \right)$.

But to find Residue at all simple poles at $z = n$ where $n \in \text{Integers}$, $(\dots, -3, -2, -1, 0, 1, 2, 3 \dots)$.

$$\begin{aligned} \text{Res}(z = n) &= \lim_{z \rightarrow n} (z - n) f(z) \\ &= \lim_{z \rightarrow n} (z - n) \cdot \frac{1}{\sin \pi z} \\ &= \lim_{z \rightarrow n} \frac{1}{\pi \cos \pi z} = \frac{(-1)^n}{\pi} \\ &\left(\text{Res}(z = n) = \frac{(-1)^n}{\pi} \right). \end{aligned}$$

(C) Multiple poles

$$f(z) = \frac{c_{-m}}{(z-a)^m} + \dots + \frac{\boxed{c_{-1}}}{(z-a)} + \underbrace{\sum_{n=0}^{\infty} c_n (z-a)^n}_{\text{Regular part.}}$$

↙ Residue (pointing to c_{-1}); ↓ ($m > 0$) (pointing to $(z-a)^m$)

$f(z)$ is said to have pole of order m at $z = a$.

But Residue is only defined for *pole of order 1 or simple pole*

$$c_{-1} = \text{Res}(z = a) = \lim_{z \rightarrow a} \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}} [(z-a)^m f(z)]$$

(D) Essential / Isolated singularity :-

Going a step further, the singular part may involve all negative Integral powers of $(z-a)$.

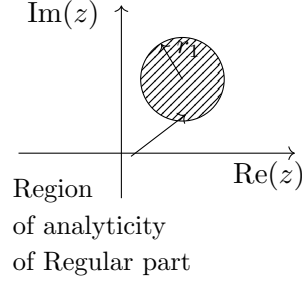
The function $f(z)$ has an isolated/essential singularity at $z = a$ if, in the neighbourhood of that point it can be expressed in form

$$f(z) = \underbrace{\sum_{n=1}^{\infty} \frac{c_{-n}}{(z-a)^n}}_{\text{Singular part.}} + \underbrace{\sum_{n=0}^{\infty} c_n (z-a)^n}_{\text{Regular part}} \quad \left\{ \begin{array}{l} \text{Laurent} \\ \text{series} \end{array} \right\}$$

regular part $\sum_{n=0}^{\infty} c_n (z-a)^n$ converges for $\left| \frac{c_{n+1}(z-a)^{n+1}}{c_n(z-a)^n} \right| < 1$

$\forall z$ which lies inside region

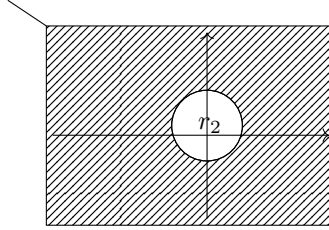
$$\begin{aligned} |z - a| < \lim_{n \rightarrow \infty} \left| \frac{c_n}{c_{n+1}} \right| & \quad |z - a| < \left| \frac{c_n}{c_{n+1}} \right| \\ (|z - a| < r_1) & \quad \text{or} \quad |z - a| < r_1 \end{aligned}$$



But the singular part, $\sum_{n=1}^{\infty} \frac{c_{-n}}{(z-a)^n}$ converges for

$$\lim_{n \rightarrow \infty} \left| \frac{c_{-n-1} (z-a)^n}{(z-a)^{n+1} c_{-n}} \right| < 1$$

Region of convergence for singular part



$$\lim_{n \rightarrow \infty} \left| \frac{1}{(z-a)} \right| \left| \frac{c_{-n-1}}{c_{-n}} \right| < 1$$

$$\begin{aligned} \text{or} \quad |z - a| & > \lim_{n \rightarrow \infty} \left| \frac{c_{-n-1}}{c_{-n}} \right| \\ (|z - a| & > r_2) \end{aligned}$$

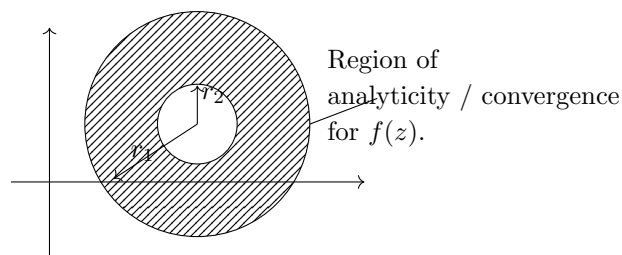
converges for,

To have representation of $f(z) = \sum_{n=1}^{\infty} \frac{c_{-n}}{(z-a)^n} + \sum_{n=0}^{\infty} c_n (z-a)^n$

in this laurent series, we need both regions to overlap.

To overlap ($r_2 < r_1$),

$$\lim_{n \rightarrow \infty} \left| \frac{c_{-n-1}}{c_{-n}} \right| < \lim_{n \rightarrow \infty} \left| \frac{c_n}{c_{n+1}} \right|$$



$$f(z) = \sum_{n=1}^{\infty} \frac{c_{-n}}{(z-a)^n} + \sum_{n=0}^{\infty} c_n (z-a)^n$$

So region of convergence becomes, $(r_2 < |z-a| < r_1)$

It is possible that $r_2 \rightarrow 0$ & $r_1 \rightarrow \infty$.

For example

(1) $f(z) = e^{\frac{1}{z}}$

$$f(z) = \underbrace{1}_{\text{Regular part}} + \underbrace{\frac{1}{z} + \frac{1}{2!} \left(\frac{1}{z}\right)^2 + \frac{1}{3!} \left(\frac{1}{z}\right)^3 + \dots}_{\text{(Singular part)}}$$

↓ ($\cdot$ Regular part) (pointing to the 1)

$$= \underbrace{\sum_{n=1}^{\infty} \frac{1}{n!} z^n}_{\text{Singular part}} + \underbrace{\boxed{1}}_{\text{Regular part}} \longrightarrow \text{converges everywhere}$$

↓

converges for

$$\lim_{n \rightarrow \infty} \left| \frac{(n)! z^n}{(n+1)! z^{n+1}} \right| < 1$$

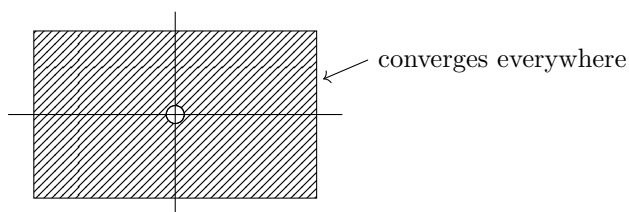
$$\lim_{n \rightarrow \infty} \left| \frac{1}{(n+1)z} \right| < 1$$

$$|z| > \lim_{n \rightarrow \infty} \left| \frac{1}{n+1} \right|$$

$$(|z| > 0) \quad \text{i.e. all } z \text{ except } \underline{(z \neq 0)} \quad z = 0.$$

$f(z)$ converges $\forall z \neq 0$. including point at infinity.

So Region of analyticity/convergence is a punctured disk.



Q) What is the residue ?

Residue is coefficient of $\frac{1}{z}$ which is 1 for $f(z) = e^{1/z}$.

$$(2) f(z) = e^{1/z} + e^z$$

$$= 1 + \sum_{n=1}^{\infty} \frac{1}{n!} \frac{1}{z^n} + \left(1 + z + \frac{z^2}{2!} + \dots \right)$$

$$= \underbrace{\sum_{n=1}^{\infty} \frac{1}{n!} \frac{1}{z^n}}_{\text{Singular part}} + 2 + \underbrace{\sum_{n=1}^{\infty} \frac{z^n}{n!}}_{\text{Regular part}}$$

↓ Converges for (singular part)

$$\lim_{n \rightarrow \infty} \left| \frac{1}{(n+1)!} \frac{1}{z^{n+1}} \cdot n! z^n \right| < 1$$

$$\lim_{n \rightarrow \infty} \left| \frac{1}{(n+1)z} \right| < 1$$

$$\text{or } |z| > \lim_{n \rightarrow \infty} \left| \frac{1}{n+1} \right|$$

$$(|z| > 0) \text{ i.e. } (\forall z \neq 0)$$

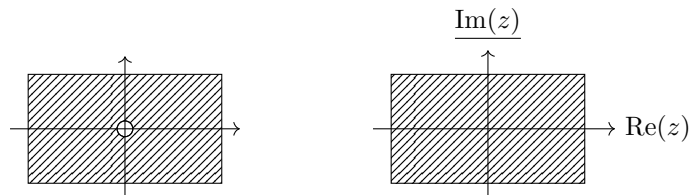
↓ Converges for (regular part)

$$\lim_{n \rightarrow \infty} \left| \frac{z^{n+1}}{(n+1)!} \frac{n!}{z^n} \right| < 1$$

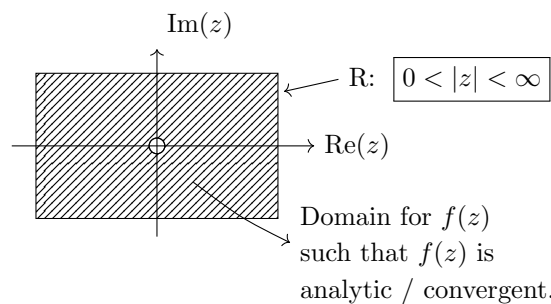
$$\lim_{n \rightarrow \infty} \left| \frac{z}{n+1} \right| < 1$$

$$|z| < \lim_{n \rightarrow \infty} |n+1|$$

$$(|z| < \infty) \text{ i.e. } \forall |z| < \infty.$$



So $f(z) = e^{1/z} + e^z$ converges for the common region



So on Riemann sphere $z = 0$ is south pole & $z = \infty$ is north pole

So $f(z)$ converges $\forall z$ except North pole & south pole.

$$\begin{aligned}
 (3) \quad f(z) &= e^{\frac{1}{z^2}} \\
 &= 1 + \frac{1}{z^2} + \frac{1}{2!} \frac{1}{z^4} + \frac{1}{3!} \frac{1}{z^6} + \dots \\
 &= \sum_{n=1}^{\infty} \frac{1}{n!} \frac{1}{z^{2n}} + \underbrace{1}
 \end{aligned}$$

↓ Converges $\forall z$ including ∞ . (under the 1)

↓

Converges for $|z| > 0$

→ $f(z)$ has essential singularity (non removable). as its singular part is summed over $\left(\sum_{n=1}^{\infty} \frac{1}{z^{2n}} \frac{1}{n!} \right)$.

→ Residue = coeff of $\frac{1}{z} = 0$.

$f(z)$ converges for $(0 < |z| < \infty)$

$$\begin{aligned}
 (4) \quad f(z) &= e^{(-)\frac{1}{(z-2)^2}} \\
 &= 1 - \frac{1}{(z-2)^2} + \left(\frac{1}{2!} \right) \cdot \frac{1}{(z-2)^4} - \frac{1}{3!} \left(\frac{1}{(z-2)^6} \right) + \dots \\
 &= \underbrace{\sum_{n=1}^{\infty} \frac{(-1)^n}{(z-2)^{2n}} \left(\frac{1}{n!} \right)}_{\text{Singular part}} + \underbrace{1}_{\text{Regular part}}
 \end{aligned}$$

So $f(z)$ has essential singularity at $z = 2$

& Residue = coeff of $\frac{1}{(z-2)} = 0$

$$\begin{aligned}
 (5) \quad f(z) &= (z-2) e^{\frac{-1}{(z-2)^2}} \\
 &= (z-2) \left(\sum_{n=1}^{\infty} \frac{(-1)^n}{(z-2)^{2n}} + 1 \right) \\
 &= (z-2) \left(1 - \frac{1}{(z-2)^2} + \frac{1}{2! (z-2)^4} \dots \right) + (z-2) \\
 &= \underbrace{\left(\frac{-1}{(z-2)} + \frac{1}{2! (z-2)^3} - \dots \right)}_{\text{Singular part}} + \underbrace{(2z-4)}_{\text{Regular part.}}
 \end{aligned}$$

It has essential singularity at $z = 2$

Residue = coeff of $\left(\frac{1}{z-2} \right) = -1$

$$(6) f(z) = (z-2)^4 e^{\frac{-1}{(z-2)^2}}$$

Again $f(z)$ involves infinite sum for singular part

So it has essential singularity.

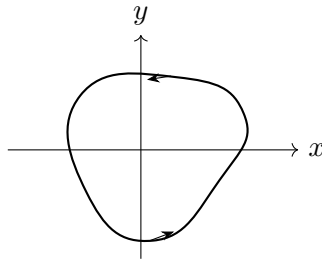
$$\begin{aligned} f(z) &= (z-2)^4 \left[1 - \frac{1}{(z-2)^2} + \frac{1}{2!} \left(\frac{1}{(z-2)^4} \right) - \frac{1}{3! (z-2)^6} + \dots \right] \\ &= \underbrace{\left((z-2)^4 - (z-2)^2 + \frac{1}{2} \right)}_{\text{Regular part}} - \underbrace{\frac{1}{3! (z-2)^2} + \dots}_{\text{Singular part.}} \end{aligned}$$

$$\text{Residue} = \text{coeff of } \left(\frac{1}{z-2} \right) = 0.$$

Cauchy's Integral formula :-

Consider integral, $\oint z^n dz$

$$I = \left(\oint z^n dz \right) =$$

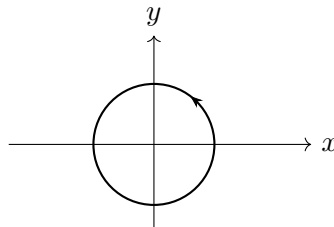


$$(n = 0, 1, 2, 3 \dots)$$

Since z^n is analytic everywhere (entire function)

$$\oint z^n dz = 0$$

we can distort the contour and shrink it to a point, or we can make it a circle $|z| = 1$.



On this unit circle $z = e^{i\theta}$ ($r = 1$)

$$dz = e^{i\theta} i d\theta$$

$$I = \oint z^n dz = \int_0^{2\pi} e^{(n+1)i\theta} i d\theta = i \int_0^{2\pi} (\cos(n+1)\theta + i \sin(n+1)\theta) d\theta$$

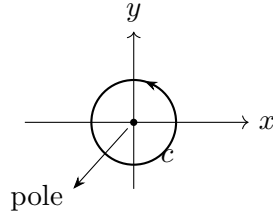
$$I = i \left(\frac{\sin(n+1)\theta}{n+1} + \frac{i \cos(n+1)\theta}{n+1} \right)_0^{2\pi} = 0$$

what happens if

$$I = \oint \frac{dz}{z^{n+1}} \quad (n = 0, 1, 2, 3 \dots)$$

$f(z) = \frac{1}{z^{n+1}}$ has pole at $z = 0$ of order $(n+1)$

since pole is at $z = 0$ and is not on contour $|z| = 1$. we can integrate again by



$$z = e^{i\theta}$$

$$dz = e^{i\theta} i d\theta$$

$$I = \oint \frac{dz}{z^{n+1}} = \int_0^{2\pi} \frac{e^{i\theta} i d\theta}{e^{(n+1)i\theta}}$$

$$I = \int_0^{2\pi} e^{-in\theta} i d\theta = \int_0^{2\pi} i (\cos n\theta - i \sin n\theta) d\theta$$

$$= i \left(\frac{\sin n\theta}{n} + \frac{i \cos n\theta}{n} \right)_0^{2\pi} = 0. \quad \forall n = 1, 2, 3 \dots$$

$$(n \neq 0)$$

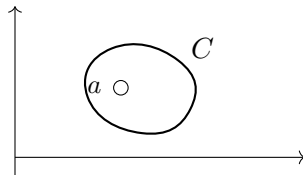
what if $(n = 0)$,

$$I = \int_0^{2\pi} i e^{-in\theta} d\theta \Big|_{(n=0)} = \int_0^{2\pi} i d\theta = 2\pi i$$

So,

$$\boxed{\frac{1}{2\pi i} \oint \frac{dz}{z^{n+1}} = \delta_{n,0}}$$

we can generalize it,



$$\left\{ \frac{1}{2\pi i} \oint_C \frac{dz}{(z-a)^{n+1}} = \delta_{n,0} \right\}$$

So, If we have a function $f(z)$, having pole at $z = a$.

Then

$$\oint f(z) dz = ?$$

(or essential singularity)

As $f(z)$ has pole, we expand $f(z)$ about $z = a$ via laurent series.

$$f(z) = \sum_{n=1}^{\infty} \frac{c_{-n}}{(z-a)^n} + \sum_{n=0}^{\infty} c_n (z-a)^n$$

$$I = \oint f(z) dz = \sum_{n=1}^{\infty} \oint \frac{dz c_{-n}}{(z-a)^n} + \sum_{n=0}^{\infty} \oint \frac{dz c_n (z-a)^n}{(z-a)^n}$$

$\hookrightarrow$ is analytic function so vanishes

$$= (2\pi i) c_{-1} + \sum_{n=2}^{\infty} \oint \frac{dz c_{-n}}{(z-a)^n} + 0$$

$$\swarrow \text{ we just saw } \frac{1}{2\pi i} \oint \frac{dz}{(z-a)^n} = \delta_{n,0}$$

$$= (2\pi i) c_{-1} + 0$$

So,

$$\left\{ \frac{1}{2\pi i} \oint f(z) dz = \text{Res}_{z=a}[f(z)] = c_{-1} \right\} \quad \begin{array}{l} \text{Cauchy} \\ \text{Integral} \\ \text{Formula} \end{array}$$

what if we find integral $\oint_C f(z) dz = ?$

here we are calculating integral over angle 0 to 4π i.e. 2 windings of contour.

So

$$\oint f(z) dz = \oint \sum_{n=1}^{\infty} \frac{c_{-n}}{(z-a)^n} + \sum_{n=0}^{\infty} \oint c_n (z-a)^n$$

$\hookrightarrow$ zero as $(z-a)^n$ is analytic.

$$\begin{aligned} &= (c_{-1}) \sum \int_0^{4\pi} \frac{dz}{(z-a)^{n+1}} + 0 \\ &= c_{-1} \int_0^{4\pi} e^{-in\theta} i d\theta + 0 \end{aligned}$$

$\checkmark$ '0' (written over the n in $-in\theta$)

$$\oint f(z) dz = (4\pi i) \text{Res}(z=a)$$

similarly for 'n' consecutive loops.

$$\oint_{(n \times c)} f(z) dz = n(2\pi i) \text{Res}(z=a)$$

$$(c \div |z| = 1)$$

If we find integral in clockwise sense

$$\begin{aligned}
 \oint f(z) dz &= \sum_{n=1}^{\infty} \oint \frac{c_{-n}}{(z-a)^n} dz + \underbrace{\sum_{n=0}^{\infty} \oint c_n (z-a)^n}_{\text{Regular part so integral}=0.} \\
 &= \oint_c \frac{c_{-1} dz}{(z-a)} + \oint \sum_{n=2}^{\infty} \frac{c_{-n} dz}{(z-a)^n} + 0 \\
 &= c_{-1} \oint \frac{dz}{(z-a)} + \sum_{n=2}^{\infty} \oint \frac{c_{-n} dz}{(z-a)^n} \\
 &\quad \begin{cases} z-a = e^{i\theta} \\ dz = e^{i\theta} i d\theta \end{cases} \\
 &= c_{-1} \int_{2\pi}^0 \frac{e^{i\theta}}{e^{i\theta}} i d\theta + \sum_{n=2}^{\infty} \oint_{2\pi}^0 \frac{c_{-n} e^{i\theta} i d\theta}{e^{in\theta}} \\
 &= c_{-1} (-2\pi i) + \sum_{n=2}^{\infty} \left(\int_{2\pi}^0 c_{-n} e^{i\theta(1-n)} i d\theta \right)
 \end{aligned}$$

$$\longrightarrow 0 \quad \forall n = 2, 3, 4 \dots$$

$$\boxed{\oint f(z) dz = -2\pi i \operatorname{Res}(z=a)}$$

In general,

$$\oint f(z) dz = n(2\pi i) \operatorname{Res}(z=a)$$

↓ no. of loops ($n \in \text{all Integers}$)

example.

$$(1) I = \oint e^{1/z} dz$$

$$\begin{aligned}
 I &= \oint \left(1 + \frac{1}{z} + \frac{1}{2! z^2} + \dots \right) dz \\
 &= \oint \frac{\boxed{1}}{z} dz + 0 \quad (\operatorname{Res}(z=0) = 1) \\
 &\quad (I = 2\pi i).
 \end{aligned}$$

$$(2) \oint e^{\frac{1}{z^2}} dz = 2\pi i \operatorname{Res}(f(z)) = 0$$

$$e^{\frac{1}{z^2}} = 1 + \frac{1}{z^2} + \frac{1}{2! z^4} + \dots$$

$$\operatorname{Res} = \left(\text{coeff of } \frac{1}{(z-0)} = \boxed{0} \right)$$